Kiarash Alirezaei

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Education

University of Toronto, St. George *Toronto, Ontario*

Bachelor of Applied Science in Computer Engineering (GPA: 3.74/4.0) *Sep 2022 – May 2027*

Minors and Certificates: Artificial Intelligence Engineering Minor, Engineering Business Minor, Cybersecurity Certificate

Relevant Coursework: Algorithms and Data Structures, Operating Systems (C), Distributed Systems (Go), Computer Networks, Machine Learning, Artificial Intelligence, Applied Deep Learning (Python), Probability & Applications, Computer Security, Digital Systems (Verilog), Digital Electronics, Computer Organization (Assembly)

Experience

University of Toronto – AI Research Projects *Sep 2026 – Present*

Applied AI Engineer – Undergraduate Research Assistant *Toronto, Ontario*

- Co-leading development of an embedded AI system that processes real-time audio and video streams on Raspberry Pi devices using signal fingerprinting and machine-learning techniques.
- Developing C++ and Python pipelines that connect on-device processing with server-side inference, cloud storage, and device-management services.
- Contributing to ML systems that translate visual interface content into interactive tactile representations for blind and low-vision users.

IBM Canada *May 2025 – Aug 2026*

Compiler Developer Intern – IBM Python for z/OS *Markham, Ontario*

- Enhanced IBM Open Enterprise SDK for Python using C and Python, integrating z/OS system functions into CPython and debugging compiler and runtime compatibility issues.
- Investigated customer-reported import, package-installation, and application issues; contributed fixes to upstream Python projects including distutils.
- Automated download, installation, and validation of IBM Python AI Toolkit packages on z/OS; authored technical articles on Python libraries, tools, installation, and debugging.
- Integrated upstream libraries such as zstd and maintained automated test and CI pipelines using Git and Buildbot to support reliable Python builds and regression testing on z/OS.

Reviewerly *May 2024 – Aug 2024*

Software Engineer Intern *Toronto, Ontario*

- Developed RESTful APIs and backend functionality using Java and MongoDB, supporting data creation, retrieval, and application workflows.
- Migrated search functionality from MongoDB queries to Apache Solr, reducing search response times by over 50%.
- Built responsive React interfaces and integrated backend APIs using React Query, improving state management, data visualization, and user interaction.

Projects

Learning-Based LLM Router (ECE496 Capstone) | *Python, ML, NLP, Model Routing* *Sep 2026 – Present*

- Designing a learned router that selects one candidate language model per prompt before inference, optimizing response quality and cost without querying every available model.
- Analyzed CrossRouter, IRT-Router, RouterDC, and other routing approaches to guide richer prompt representations, per-model quality and cost prediction, and uncertainty-aware selection.
- Defined an offline RouterArena evaluation plan across 10 datasets using accuracy, cost, optimal-selection, robustness, and latency metrics at an accuracy-weighted $\beta = 0.1$.

KernelPilot – RMSNorm CUDA Optimization | *CUDA C++, PyTorch, GPU Kernels* *Oct 2026 – Present*

- Developing a CUDA C++ optimization project centered on RMSNorm, beginning with a baseline custom kernel and correctness comparisons against PyTorch CPU and GPU implementations.
- Planning Nsight profiling, block and thread tuning, memory-access optimization, fused RMSNorm–GELU variants, and automated kernel-configuration selection on an NVIDIA RTX 5060 Ti.

Projects (continued)

Fault-Tolerant Distributed Key-Value Store | *Go, Raft, RPC, Distributed Systems* *Jan 2025 – Apr 2025*

- Implemented the Raft consensus protocol in Go, including randomized leader election, heartbeat-based failure detection, replicated logs, majority-based commits, and automatic recovery from node failures.
- Developed persistent state and optimized log-reconciliation mechanisms to recover crashed replicas, resolve conflicting logs, and efficiently resynchronize nodes after network partitions.
- Built a linearizable distributed key-value store on top of Raft, supporting concurrent Get/Put operations, client retries, leader changes, server restarts, and unreliable network conditions.

Skin Lesion Classifier (NLC7) | *Python, PyTorch, U-Net, ResNet-50, MLP* *May 2024 – Aug 2024*

- Processed the HAM10000 dataset with augmentation and class-balancing workflows using Pandas and PyTorch, preparing approximately 6,000 images across lesion types for model training.
- Designed a deep-learning pipeline combining U-Net segmentation, ResNet-50 transfer learning for feature extraction, and an MLP classifier for skin-lesion classification.
- Researched image-processing techniques and tuned U-Net, CNN, and MLP models, achieving approximately 92% accuracy while reducing pipeline RAM usage by approximately 75%.

Interactive Maps & Routing Engine | *C++ (STL), EZGL, GTK* *Feb 2024 – Mar 2024*

- Developed an interactive C++ mapping application with real-time data APIs, a responsive interface, and route optimization using A*, multi-source Dijkstra's algorithm, and a Traveling Salesman Problem heuristic.

Digital Game System – FPGA Development | *Verilog, ModelSim, DE1-SoC, VGA, PS/2* *Sep 2023 – Dec 2023*

- Engineered a digital game system in Verilog on a DE1-SoC FPGA, integrating finite-state machines, datapaths, memory blocks, VGA output, PS/2 input, and audio functionality.
- Designed and verified control logic and reusable memory components in ModelSim before deploying the system to the FPGA development board.

Technical Skills

Languages: Python, C, C++, CUDA C++, Java, Go, JavaScript, SQL, Bash/Shell, Verilog, Assembly, MATLAB
Systems & Platforms: CPython, z/OS USS, Linux, POSIX APIs, Embedded Systems, Raspberry Pi, Compiler & Runtime Development, Distributed Systems, Concurrency, Fault Tolerance, RPC
AI & Machine Learning: PyTorch, TensorFlow, NumPy, Pandas, Deep Learning, Computer Vision, CNNs, U-Net, ResNet, LLM Routing, NLP, Model Selection, Signal Fingerprinting
Frameworks & Software: React, React Native, Spring Boot, Django, REST APIs, gRPC, GTK, EZGL
Hardware & GPU: CUDA, FPGA (DE1-SoC), Quartus, ModelSim, VGA, PS/2, Finite-State Machines, Datapaths, Digital Logic
Cloud & DevOps: Azure, AWS, Git, Buildbot, Ansible, Docker, CI/CD, pytest, Test Automation, Python Packaging, pip, Bandit, Detect-Secrets
Databases & Search: MongoDB, MySQL, Apache Solr